

binomial distribution
is a discrete
probability distribution

$$P(x) = \binom{n}{x} p^x q^{n-x}$$

0/1 0/1 0/1 . . . —
n dashes.

We have exactly x 1s and $n-x$ 0s in
specific places

1 — 1 — 1 . . . —

x 1s in specific positions
 $n-x$ 0s in the remaining positions

$p^x q^{n-x}$ is the probability that
 n.i.

$p^x q^{n-x}$ in the probability
 there are x 1s and $n-x$ 0s in
 specific positions

$\binom{n}{x}$ is the number of ways in which
 we can pick out x positions out of n
 when we need to place the 1s.

$$\binom{n}{x} p^x q^{n-x}$$

$$\frac{11001010}{p^x q^{n-x}} + \frac{11110000}{p^x q^{n-x}} + \frac{10010010}{p^x q^{n-x}}$$

$$\binom{n}{x} p^x q^{n-x}$$

$\binom{n}{0} p^0 q^{n-0} \rightarrow$ probability of getting 0 1s

$\binom{n}{1} p^1 q^{n-1} \rightarrow$ probability of getting 1 1

$$\binom{n}{0} p^0 q^{n-0} + \binom{n}{1} p^1 q^{n-1} + \dots + \binom{n}{n} p^n q^{n-n} = 1$$

$(p+q)^n$ = Left hand side is a binomial expansion of $(p+q)^n = 1^n = 1$.

if X has a binomial distribution what is its expectation

$$E(X) = np?$$

$X = Y_1 + Y_2 + \dots + Y_n$ when Y_i is a Bernoulli random variable independent of all the others

$$E(X) = E(Y_1 + Y_2 + \dots + Y_n) = E(Y_1) + E(Y_2) + \dots + E(Y_n)$$

$$E(Y_1) = E(Y_2) = \dots = E(Y_n) = p$$

$$\therefore \boxed{E(X) = np}$$

$$\text{Var}(X) = \text{Var}(Y_1 + Y_2 + \dots + Y_n)$$

Since $Y_1, Y_2, \dots, Y_n$ are all independent random variables

$$\text{Var}(Y_1 + Y_2 + \dots + Y_n) = \text{Var}(Y_1) + \text{Var}(Y_2) + \dots + \text{Var}(Y_n)$$

$$\text{Var}(Y_1) = p - p^2 = \underbrace{E(Y_1^2)}_p - \underbrace{[E(Y_1)]^2}_{p^2} = p(1-p)$$

$$\text{Var}(X) = np(1-p)$$

$$\sum_{x=0}^n x \binom{n}{x} p^x q^{n-x} = np. \quad \left[\text{You can do it this way too!} \right]$$

$$\sum_{x=0}^n x^2 \binom{n}{x} p^x q^{n-x} =$$

$$\sum p(x) = \sum e^{-\lambda} \frac{\lambda^x}{x!} = e^{-\lambda} \left(\sum_{x=0}^{\infty} \frac{\lambda^x}{x!} \right)$$

e^{λ}

$$e^{-\lambda} \cdot e^{\lambda} = e^0 = 1$$

$$\binom{n}{x} p^x q^{n-x} \xrightarrow{1} e^{-\lambda} \frac{\lambda^x}{x!}$$

$$\frac{n!}{(n-x)! x!} = \frac{n(n-1)(n-2) \dots [n-(x+1)] (n-x)!}{(n-x)! x!}$$

$$\frac{n(n-1)(n-2) \dots (n-x+1)}{1 \cdot 2 \cdot 3 \dots x}$$

$$\frac{n \cdot n(1 - \frac{1}{n}) \cdot n(1 - \frac{2}{n}) \dots n(1 - \frac{x-1}{n})}{1 \cdot 2 \cdot \dots \cdot x}$$

$$\frac{n^x}{x!} \left(1 - \frac{1}{n}\right) \left(1 - \frac{2}{n}\right) \dots \left(1 - \frac{x-1}{n}\right)$$

$$\binom{n}{x} \approx \frac{n^x}{x!}$$

$$\binom{n}{x} p^x (1-p)^{n-x} = \frac{n^x}{x!} p^x (e^{-p})^{n-x}$$

$$= \frac{n^x}{x!} p^x e^{-np} \left(e^{\frac{p^x}{n}} \right) \approx 1$$

$$= \frac{(np)^x}{x!} e^{-np}$$

$$= \frac{e^{-\lambda} \lambda^x}{x!} = \frac{e^{-\lambda} \lambda^x}{x!}$$

Binomial Distribution

$$E(X) = np$$

$$\text{Var}(X) = np(1-p)$$

Poisson distribution

$$n \rightarrow \infty, p \rightarrow 0$$

$$E(X) = np$$

$$\text{Var}(X) = np - \frac{np^2}{(np)p} \rightarrow 0$$

$$\frac{\text{Var}(X) = np}{E(X) = np}$$

X follows $N(250, 275)$

$$P(X \geq 15) ?$$

$$\int_{15}^{\infty} \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}$$

Convert it to a standard normal distribution

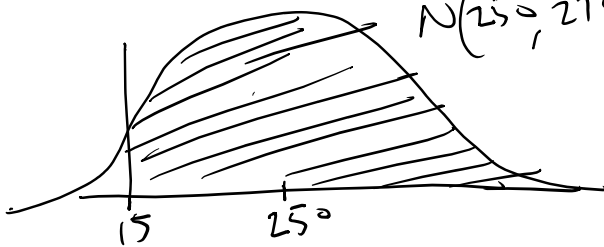
$$\frac{15 - \mu}{\sigma} = \frac{15 - 250}{275}$$

$$Y \geq \frac{15 - 250}{275}$$

original dist

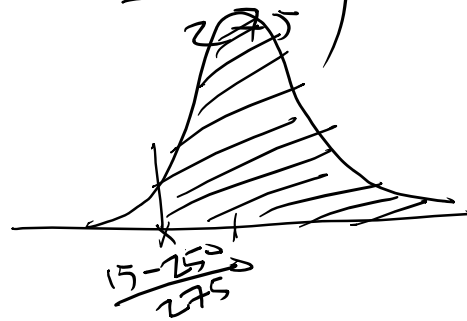
$$P(X \geq 15)$$

$N(250, 275)$

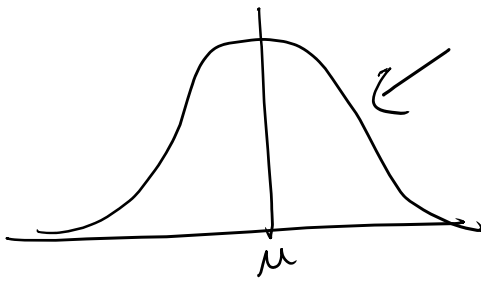


$$P\left(Y \geq \frac{15 - 250}{275}\right)$$

in the std normal dist

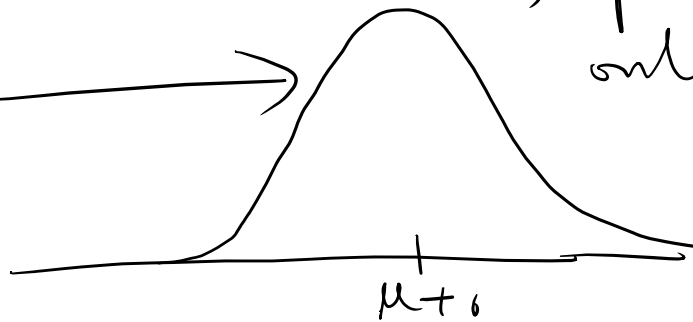


$$Y = X + 1$$



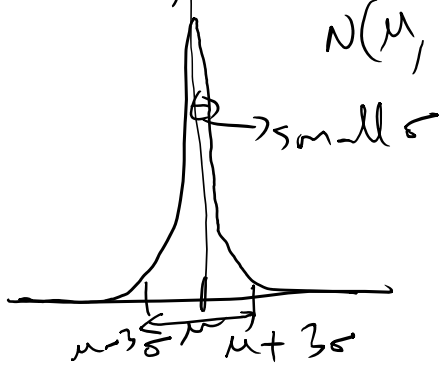
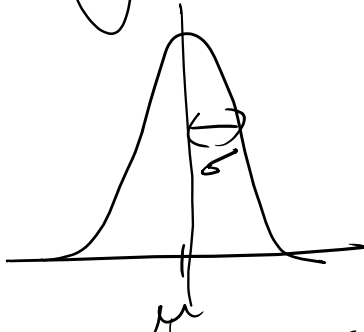
distribution of X

$$Y = X + 1$$

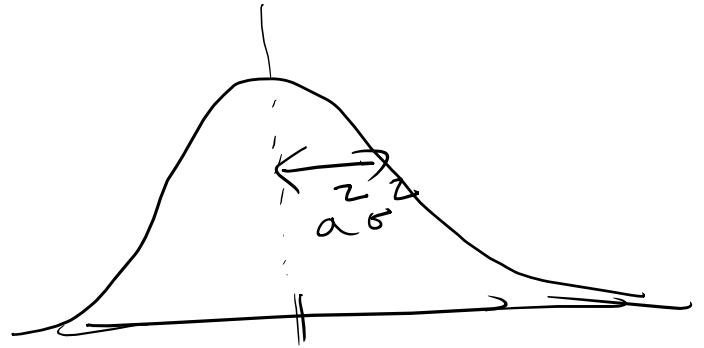


shape stays the same,
only mean
shifted

$$Y = \textcircled{a}X + b$$

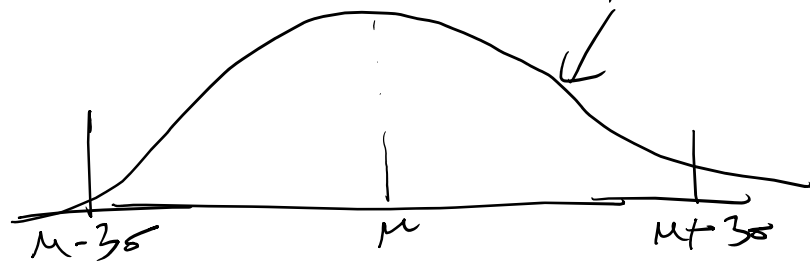


$N(\mu, \text{small } \sigma)$



$a\mu + b$

$N(\mu, \text{large } \sigma)$



$$\bar{X} = \frac{\sum X_i}{n} \quad X_i \rightarrow \text{normal dist}$$

$$\frac{\sum X_i}{n} \rightarrow \text{also normal}$$

$$\bar{X} - \mu \rightarrow \text{normal dist. lution}$$

$$S = \sqrt{\frac{1}{(n-1)} \sum_{i=1}^n (X_i - \bar{X})^2}$$

$$\text{degrees of freedom} = \underline{(n-1)}, \text{ not } n$$

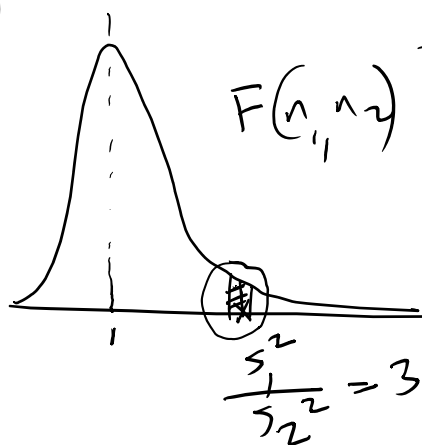
When is the F-distribution actually used?

Comparing two variances.

Question: Are the variances of the two populations equal?

Procedure: $F = \frac{s_1^2}{s_2^2}$ when s_1^2 and s_2^2

are the two sample variances, and we place the larger sample variance in the numerator



highly unlikely that $s_1^2 = 3s_2^2$ if the original populations had the same variance

null hypothesis: $\sigma_1^2 = \sigma_2^2$